IDC

ANALYST CONNECTION

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Enterprises that prioritize unified governance, decentralized access, robust data readiness, and measurable outcomes are best positioned to realize the full potential of AI.

AI and Data Modernization: Enterprise Readiness and Value Realization

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Questions posed by: Qlik and Artha Solutions

Answers by: Stewart Bond, Research Vice President, Data Intelligence and Integration Software

Q. How are enterprises reacting to AI, and are leaders aligned with business objectives for AI adoption?

A. AI's rapid evolution is prompting enterprises to move decisively from experimentation to adoption. This shift is not only technological but also organizational, as enterprises seek to realize business value from their AI investments. The primary business objectives driving AI adoption are increased operational efficiency (including cost savings), faster innovation, and revenue growth. These priorities reflect a clear recognition that AI is a lever for both productivity and competitive differentiation.

Unfortunately, not all AI projects deliver measurable business value, according to IDC's research. Only 45% of AI projects demonstrate business value, with competition for resources, resistance to process changes, and regulatory uncertainty topping the list of factors challenging the realization of AI's potential. A critical factor in AI's successful adoption is the alignment between IT and business leadership, so that resources are balanced, change management is implemented, and regulatory compliance requirements are well understood. A significant proportion of AI project sponsors now come from employees in non-IT roles. This highlights the importance of cross-functional collaboration and underscores the need for organizations to ensure that AI initiatives are closely tied to business outcomes.

Enterprises are prioritizing AI integration in various domains, including IT, customer service, operations, and manufacturing. These areas are seen as fertile grounds for quick wins, as they offer clear opportunities for agentic AI to drive measurable improvements in efficiency and responsiveness, enabling momentum and demonstrating value.

As enterprises continue to respond to AI's opportunities and challenges, assuring data is ready for AI model training, tuning, and use at the time of inference is now a priority. If data isn't production ready, AI solutions will also not be production ready. Establishing a unified data foundation is the first step to AI success. At the beginning of 2025, an IDC survey of IT professionals worldwide indicated that AI infrastructure was the most important area of focus to enable organizations to move AI into production. By midyear, another IDC survey of IT professionals showed that ensuring data quality and availability for AI models was now the highest priority. This shift suggests that either all of the infrastructure problems have been resolved, which is unlikely, or that organizations have experienced failed initiatives because the data was not AI ready.

Q. What are the fastest, most objective ways to diagnose AI data readiness, and what early indicators prove the foundation is fit or fragile?

A. Assessing AI data readiness is a foundational step for organizations aiming to scale AI initiatives and achieve reliable outcomes. The most effective approaches are those that provide objective, actionable insights into the current state of data infrastructure, governance, and quality.

Early signs that a data foundation may be fragile include persistent issues with data intelligence. These include a lack of data quality and related metrics, as well as a lack of cataloging, lineage, and semantics, which can cause insufficient governance mechanisms. These weaknesses often emerge as barriers to scaling AI. They suggest that improving data intelligence is the most important investment organizations can make in getting data AI ready. Organizations that have mature and robust capabilities in harvesting, leveraging, and monitoring data intelligence KPIs have demonstrated better business outcomes, such as improving innovation and time to market for new product metrics by 8% and improving customer acquisition and retention by 7%. These organizations also saw an average improvement in profit and revenue by 7% (source: IDC's *Office of the CDO Survey,* August 2024).

Unified data platforms and federated architectures cannot exist without data intelligence, and these architectures play a pivotal role in enabling real-time data accessibility, which is essential for supporting dynamic AI use cases. These architectures facilitate faster joining of data sets, improve data quality, and support the agility required for agentic AI. End-to-end observability across the data value chain enhances transparency, accelerates the transition from proof of concept to production, and increases the success rate of AI deployments. Every chief data officer (or equivalent) needs to demonstrate the business value of data; not being able to make this connection, every AI decision is suspect.

Uncovering technology challenges is only part of the assessment. Organizational issues can also undermine AI initiatives and data readiness. Data leaders face a number of challenges when it comes to AI, starting with managing expectations about what it can deliver. Other concerns include dealing with skills development and resource constraints, managing changing priorities, orchestrating collaboration between business and IT, and addressing budget constraints. Best practices, including proper change management, should be put in place to help address technical and organizational challenges.

Q. What are the best practices for achieving data readiness and maturity for AI, and why is this foundational to AI success?

A. Achieving data readiness and maturity is a multifaceted challenge that requires both technical and organizational strategies. Issues of data security and governance, quality and availability, coupled with the organizational challenges of managing expectations, competing for resources, and enabling collaboration, can be magnified in modern data environments, in which data is increasingly distributed, diverse, dynamic, and dark.

Best practices for overcoming these barriers begin with treating data as a product, rather than a by-product. Historically, data has been treated as a by-product of applications, and it has been everybody's responsibility, which means that it is nobody's responsibility. Treating data as a product requires assigning clear ownership, curating data for quality and relevance, and aligning data management practices with business outcomes. Organizations can accelerate time to value, foster innovation, and ensure that data is both accessible and trustworthy for AI applications.

Implementing robust governance frameworks fueled by data intelligence is another essential practice. Effective governance ensures compliance, maintains data integrity, and supports the scalability of AI initiatives. Equally important is a focus on data literacy and change management for both technical and business users. Without a concerted effort to build skills and drive adoption, even the most advanced data solutions may fail to deliver their full potential.

External service providers can play a critical role in bridging skill gaps and connecting services to help deliver measurable business value. Organizations investing in enterprise intelligence services report significant gains in business KPIs, on average, including 24% faster innovation, 23% increase in operational efficiency, 23% improvement in business agility, 21% revenue growth, and 19% cost savings (source: IDC's *Enterprise Intelligence Services Survey,* June 2025). These outcomes highlight the foundational role of data readiness in achieving AI success and underscore the importance of partnering with a services provider for assistance with architecture and delivering a holistic, best practice–driven approach.

Q. How should organizations approach data architecture, governance, and KPIs to maximize their AI outcomes?

A. The architecture of data environments is evolving rapidly to meet the demands of agentic AI. The prevailing trend is toward decentralized data access combined with centralized governance. This approach enables organizations to maintain compliance and control while empowering domain-specific teams with the flexibility to access and utilize data as needed. By 2027, most agentic AI use cases are expected to require federated data models, according to IDC, reflecting the growing importance of distributed and federated data architectures.

Centralized governance frameworks are essential for maintaining oversight and ensuring that data policies are consistently applied across the organization. At the same time, decentralized domain ownership and the use of data products to improve data access controls help avoid bottlenecks and the so-called "data tax." This is the hidden cost in time, effort, risk, and lost value that arises when organizations impose overly centralized, rigid, or bureaucratic controls over how data is accessed, shared, governed, or used. Event-driven architectures that leverage change data capture and streaming technologies are also being adopted to support real-time AI processing, controlled distribution, and faster access, further enhancing agility and responsiveness. Leveraging data virtualization and open table formats can also help decentralize data ownership and improve data utility.

KPIs for governance should encompass data quality, lineage, privacy, and observability. Organizations that have established effective data management technologies and practices see higher levels of improvement in data and analytics quality, data security and privacy, availability of data for AI, and data worker productivity KPIs. These metrics have a direct impact on AI and business outcomes, where we see average operational improvements of 6% with time to market, innovation, and customer retention metrics being above the average. These organizations are also experiencing an average improvement of 4% in financial metrics, with profit and revenue being above the average (source: IDC's *Office of the CDO Survey,* August 2024).

Q. What practical steps should organizations take today to prepare for the next wave of AI innovation?

A. Organizations need to take an incremental approach to build a strong foundation within the first 90 days, then scale and optimize AI capabilities and governance over the following 9 months, positioning them for success in the next wave of AI innovation. In detail:

- **The first 90 days — foundation and assessment:**
  - Form a cross-functional AI oversight group or center of excellence with business and IT leaders.
  - Conduct an AI-/data-readiness assessment to identify gaps in data quality, governance, and infrastructure.
  - Develop and communicate basic risk-based AI governance guidelines, focusing on transparency, data integrity, and ethical use.
  - Inventory current AI initiatives and data assets, prioritizing quick wins and areas for improvement.
  - Initiate regular collaboration meetings between business and technology stakeholders to align on AI strategy and governance.
- **The next nine months — build, scale, and optimize:**
  - Expand the AI oversight group's role to include developing a comprehensive AI road map and aligning governance policies with business strategy and ethics.
  - Implement a unified data governance framework to support multiple AI use cases, ensuring scalability and compliance.
  - Launch pilot AI projects with clear KPIs, leveraging lessons learned to refine governance and operational practices.
  - Establish continuous auditing and controls for AI outputs to monitor for bias, drift, and compliance issues.
  - Provide ongoing education and training for staff on AI risks, opportunities, and responsible use.
  - Foster strong partnerships with technology providers and external experts to bridge capability gaps and accelerate innovation.
  - Review and update governance policies and practices to adapt to evolving AI technologies and regulatory requirements.

About the Analyst

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Stewart Bond is research vice president of IDC's Data Intelligence and Integration Software service. Mr. Bond's core research coverage includes watching emerging trends that are shaping and changing data movement, ingestion, transformation, mastering, cleansing, and consumption in the era of digital business. Having worked in the IT industry for over 30 years, from early experience in database and application development, through solution design and deployment, to strategic architectural consulting, Stewart has worked through some significant changes in the IT industry.

MESSAGE FROM THE SPONSOR

Data is growing fast and spreading across apps, clouds, and teams. Without clear control, it turns into silos, extra cost, and AI that never leaves pilot.

Artha Solutions Data Foundation Services make data ready for AI. Artha sets the operating model — catalog, lineage, quality, privacy, and always-on monitoring — while Qlik provides the integration backbone for CDC/replication, ELT, and federation. Together, they connect the systems that create and use data so people get trusted, governed access to both structured and unstructured information.

With Artha, customers can:

» Ship AI faster — Clean, connected data and reusable building blocks turn pilots into live products.

» Trust every decision — See where data came from, keep it fresh, and catch drift or bias before it reaches downstream.

» Scale without surprises — Use open standards and move compute to the data to cut hidden costs and avoid lock-in.

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